

**G5 Implicit differentiation
and parametric
differentiation A level only**

Name: _____

Class: _____

Date: _____

Time: **300 minutes**

Marks: **250 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A curve is defined by the parametric equations

$$x = 4 \times 2^{-t} + 3$$

$$y = 3 \times 2^t - 5$$

(a) Show that $\frac{dy}{dx} = -\frac{3}{4} \times 2^{2t}$

(3)

(b) Find the Cartesian equation of the curve in the form $xy + ax + by = c$, where a , b and c are integers.

(3)

(Total 6 marks)

Q2.

Find the coordinates of the stationary point of the curve with equation

$$(x + y - 2)^2 = e^y - 1$$

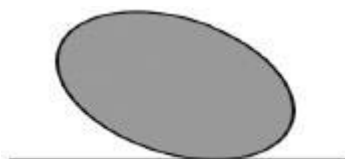
(Total 7 marks)

Q3.

A sculpture formed from a prism is fixed on a horizontal platform, as shown in the diagram.

The shape of the cross-section of the sculpture can be modelled by the equation $x^2 + 2xy + 2y^2 = 10$, where x and y are measured in metres.

The x and y axes are horizontal and vertical respectively.



Find the maximum vertical height above the platform of the sculpture.

(Total 8 marks)

Q4.

A curve is defined by the parametric equations

$$x = t^3 + 2, \quad y = t^2 - 1$$

(a) Find the gradient of the curve at the point where $t = -2$

(4)

- (b) Find a Cartesian equation of the curve.

(2)

(Total 6 marks)

Q5.

- (a) A curve is defined by the equation $x^2 - y^2 = 8$.

- (i) Show that at any point (p, q) on the curve, where $q \neq 0$, the gradient of the curve is given by $\frac{dy}{dx} = \frac{p}{q}$.

(2)

- (ii) Show that the tangents at the points (p, q) and $(p, -q)$ intersect on the x -axis.

(4)

- (b) Show that $x = t + \frac{2}{t}$, $y = t - \frac{2}{t}$ are parametric equations of the curve $x^2 - y^2 = 8$.

(2)

(Total 8 marks)

Q6.

A curve is defined by the parametric equations $x = 8e^{-2t} - 4$, $y = 2e^{2t} + 4$.

- (a) Find $\frac{dy}{dx}$ in terms of t .

(3)

- (b) The point P , where $t = \ln 2$, lies on the curve.

- (i) Find the gradient of the curve at P .

(1)

- (ii) Find the coordinates of P .

(2)

- (iii) The normal at P crosses the x -axis at the point Q . Find the coordinates of Q .

(3)

- (c) Find the Cartesian equation of the curve in the form $xy + 4y - 4x = k$, where k is an integer.

(3)

(Total 12 marks)

Q7.

A curve is defined by the parametric equations

$$x = 8t^2 - t, \quad y = \frac{3}{t}$$

- (a) Show that the cartesian equation of the curve can be written as $xy^2 + 3y = k$, stating the value of the integer k . (2)

- (b) (i) Find an equation of the tangent to the curve at the point P , where $t = \frac{1}{4}$. (7)

- (ii) Verify that the tangent at P intersects the curve when $x = \frac{3}{2}$. (2)

(Total 11 marks)

Q8.

A curve is defined by the parametric equations

$$x = 2 \cos \theta, \quad y = 3 \sin 2\theta$$

- (a) (i) Show that

$$\frac{dy}{dx} = a \sin \theta + b \operatorname{cosec} \theta$$

where a and b are integers. (4)

- (ii) Find the gradient of the normal to the curve at the point where $\theta = \frac{\pi}{6}$. (2)

- (b) Show that the cartesian equation of the curve can be expressed as

$$y^2 = px^2(4 - x^2)$$

where p is a rational number. (3)

(Total 9 marks)

Q9.

A curve is defined by the equation $9x^2 - 6xy + 4y^2 = 3$.

Find the coordinates of the two stationary points of this curve.

(Total 8 marks)

Q10.

A curve is defined by the parametric equations

$$x = 3e^t, \quad y = e^{2t} - e^{-2t}$$

- (a) (i) Find the gradient of the curve at the point where $t = 0$. (3)

- (ii) Find an equation of the tangent to the curve at the point where $t = 0$. (1)

- (b) Show that the cartesian equation of the curve can be written in the form

$$y = \frac{x^2}{k} - \frac{k}{x^2}$$

where k is an integer.

(2)

(Total 6 marks)

Q11.

- (a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

- (i) Show that $\frac{dy}{dx} = \frac{1}{k \cos \theta}$, where k is an integer. (4)

- (ii) Find an equation of the normal to the curve at the point where $\theta = \frac{\pi}{3}$. (4)

- (b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$. (5)

(Total 13 marks)

Q12.

A curve is defined by the equation $2y + e^{2x}y^2 = x^2 + C$, where C is a constant.

The point $P = \left(1, \frac{1}{e}\right)$ lies on the curve.

- (a) Find the exact value of C . (1)

- (b) Find an expression for $\frac{dy}{dx}$ in terms of x and y . (7)

- (c) Verify that $P = \left(1, \frac{1}{e}\right)$ is a stationary point on the curve.

(2)

(Total 10 marks)

Q13.

A curve is defined by the equation

$$x^2 + xy = e^y$$

Find the gradient at the point $(-1, 0)$ on this curve.

(Total 5 marks)

Q14.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$.

(2)

- (ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$.

(2)

- (b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ .

(3)

- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P .

(3)

(Total 10 marks)

Q15.

A curve is defined by the parametric equations

$$x = 1 - 3t, \quad y = 1 + 2t^3$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(3)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Find a cartesian equation of the curve.

(2)

(Total 9 marks)

Q16.

A curve has equation $x^3y + \cos(\pi y) = 7$.

- (a) Find the exact value of the x -coordinate at the point on the curve where $y = 1$.

(2)

- (b) Find the gradient of the curve at the point where $y = 1$.

(5)

(Total 7 marks)

Q17.

A curve is defined by the equation $x^2y + y^3 = 2x + 1$.

- (a) Find the gradient of the curve at the point $(2, 1)$.

(6)

- (b) Show that the x -coordinate of any stationary point on this curve satisfies the equation

$$\frac{1}{x^3} = x + 1$$

(4)

(Total 10 marks)

Q18.

A curve is defined by the parametric equations

$$x = \frac{1}{t}, \quad y = t + \frac{1}{2t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(4)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Show that the cartesian equation of the curve can be written in the form

$$x^2 - 2xy + k = 0$$

where k is an integer.

(3)

(Total 11 marks)

Q19.

A curve is defined by the equation $4x^2 + y^2 = 4 + 3xy$.

Find the gradient at the point $(1, 3)$ on this curve.

(Total 5 marks)

Q20.

A curve is defined by the parametric equations $x = 2t + \frac{1}{t^2}$, $y = 2t - \frac{1}{t^2}$.

(a) At the point P on the curve, $t = \frac{1}{2}$.

(i) Find the coordinates of P .

(2)

(ii) Find an equation of the tangent to the curve at P .

(5)

(b) Show that the cartesian equation of the curve can be written as

$$(x - y)(x + y)^2 = k$$

where k is an integer.

(3)

(Total 10 marks)

Q21.

A curve has equation $3xy - 2y^2 = 4$.

Find the gradient of the curve at the point $(2, 1)$.

(Total 5 marks)

Q22.

A curve is defined, for $t \neq 0$, by the parametric equations

$$x = 4t + 3, \quad y = \frac{1}{2t} - 1$$

At the point P on the curve, $t = \frac{1}{2}$.

- (a) Find the gradient of the curve at the point P . (4)
- (b) Find an equation of the normal to the curve at the point P . (3)
- (c) Find a cartesian equation of the curve. (3)
- (Total 10 marks)**

Q23.

A curve is defined by the parametric equations

$$x = 1 + 2t, \quad y = 1 - 4t^2$$

- (a) (i) Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (2)
- (ii) Hence find $\frac{dy}{dx}$ in terms of t . (2)
- (b) Find an equation of the normal to the curve at the point where $t = 1$. (4)
- (c) Find a cartesian equation of the curve. (3)
- (Total 11 marks)**

Q24.

The point $P(1, a)$, where $a > 0$, lies on the curve $y + 4x = 5x^2y^2$.

- (a) Show that $a = 1$. (2)
- (b) Find the gradient of the curve at P . (7)
- (c) Find an equation of the tangent to the curve at P . (1)
- (Total 10 marks)**

Q25.

A curve is given by the parametric equations

$$x = \cos \theta \quad y = \sin 2\theta$$

- (a) (i) Find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ (2)

- (ii) Find the gradient of the curve at the point where $\theta = \frac{\pi}{6}$ (2)

- (b) Show that the cartesian equation of the curve can be written as

$$y^2 = kx^2(1 - x^2)$$

where k is an integer.

(4)

(Total 8 marks)

Q26.

A curve is defined by the parametric equations

$$x = 3 - 4t \quad y = 1 + \frac{2}{t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t . (4)

- (b) Find the equation of the tangent to the curve at the point where $t = 2$, giving your answer in the form $ax + by + c = 0$, where a , b and c are integers. (4)

- (c) Verify that the cartesian equation of the curve can be written as

$$(x - 3)(y - 1) + 8 = 0$$

(3)

(Total 11 marks)

Q27.

A curve is defined by the equation

$$y^2 - xy + 3x^2 - 5 = 0$$

- (a) Find the y -coordinates of the two points on the curve where $x = 1$. (3)

- (b) (i) Show that $\frac{dy}{dx} = \frac{y - 6x}{2y - x}$. (6)

- (ii) Find the gradient of the curve at each of the points where $x = 1$.

(2)

- (iii) Show that, at the two stationary points on the curve, $33x^2 - 5 = 0$.

(3)

(Total 14 marks)

Q28.

A curve is defined by the parametric equations

$$x = 2t + \frac{1}{t}, \quad y = \frac{1}{t}, \quad t \neq 0$$

- (a) Find the coordinates of the point on the curve where $t = \frac{1}{2}$.

(2)

- (b) Show that the cartesian equation of the curve can be written as

$$xy - y^2 = 2$$

(2)

- (c) Show that the gradient of the curve at the point (3, 2) is 2.

(6)

(Total 10 marks)